

Methods of differentiation, critical points, and optimization

The product rule: Let u and v be functions of x . Then

$$\frac{d}{dx}uv = v\frac{du}{dx} + u\frac{dv}{dx}$$

Example: Let

$$f(x) = (1 - 20x^2)\left(\frac{1}{x} + 2\right)$$

Find $f'(x)$

The quotient rule: Let u and v be functions of x . Then

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \left(v \frac{du}{dx} - u \frac{dv}{dx} \right) / v^2$$

Or: “Low D high minus high D low, square the bottom and away we go!”

Example: Let

$$f(x) = \frac{3x + a}{x^2 + b}$$

Find $f'(x)$

The composite function rule (aka the chain rule): Let $f(x) = g(h(x))$.
Then

$$f'(x) = g'(h(x)) \cdot h'(x)$$

Example 1: Find $f'(x)$ for

$$f(x) = \left(\frac{5}{x} + 2x^2\right)^2$$

Example 2: Find $f'(x)$ for

$$f(x) = \frac{1}{g(x)}$$

Does this look familiar?

An example with voter preferences

Suppose that a voter has ideal point $v \in \mathbb{R}$, and receives utility from a policy x equal to

$$u(x) = -(v - x)^2$$

Now suppose that there is a politician with ideal point p , and that the politician chooses a policy that is the average of his own ideal point and his party's platform, P , so that policy $x = \frac{p+P}{2}$. How does the voter's utility change as a function of the candidate's ideal point, p ? (In other words, find $\frac{du}{dp}$)

Monotonic functions: functions that are either *strictly increasing* or *strictly decreasing*

Strictly increasing: If $x_2 > x_1$ then $f(x_2) > f(x_1)$

Strictly decreasing: If $x_2 > x_1$ then $f(x_2) < f(x_1)$

If $f'(x) > 0$ for all x , then f is strictly increasing;

however this is stronger than we need for a function to be strictly increasing (e.g. $f(x) = x^3$)

If $f'(x) \geq 0$ for all x and $f'(x) = 0$ for only a finite number of x , then f is strictly increasing (with strictly decreasing proved similarly)

Inverse functions

Let $f(x)$ be monotonic. Then for any y , $f(x) = y$ has at most one solution. If the solution is written $x = g(y)$, then g is the *inverse function* of f , or

$$f(g(y)) = y \text{ and } g(f(x)) = x$$

Example 1: Let $f(x) = x^3$; what is the inverse function of f ?

Example 2: Let $f(x) = x^{-1} + 5$; what is the inverse function of f ?

The inverse function is a reflection of the function about the 45° line; what's the inverse of the function $\{(1, 2), (2, 3), (3, 4)\}$?

The inverse function rule

If f is monotonic with inverse function g , and for a given x , f is differentiable and $f'(x) \neq 0$, then

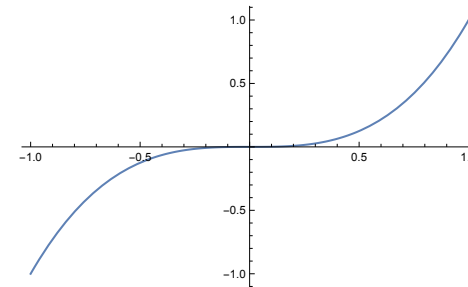
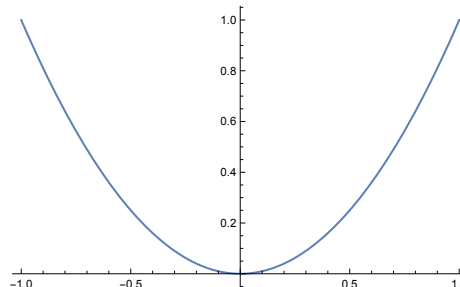
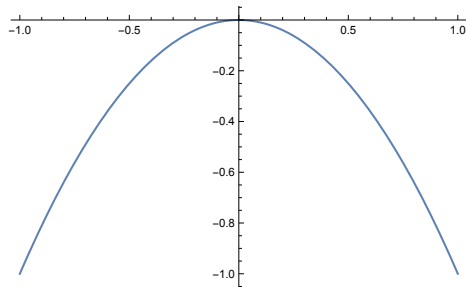
$$g'(y) = \frac{1}{f'(x)}$$

This gives us another technique for differentiating things; we could calculate f' directly, or we could calculate $\frac{1}{g'}$

Critical points

If $f'(x) = 0$ then x is called a *critical point*

- Maximum
- Minimum
- Inflection point



One of the main things we use calculus for is to identify and classify critical points

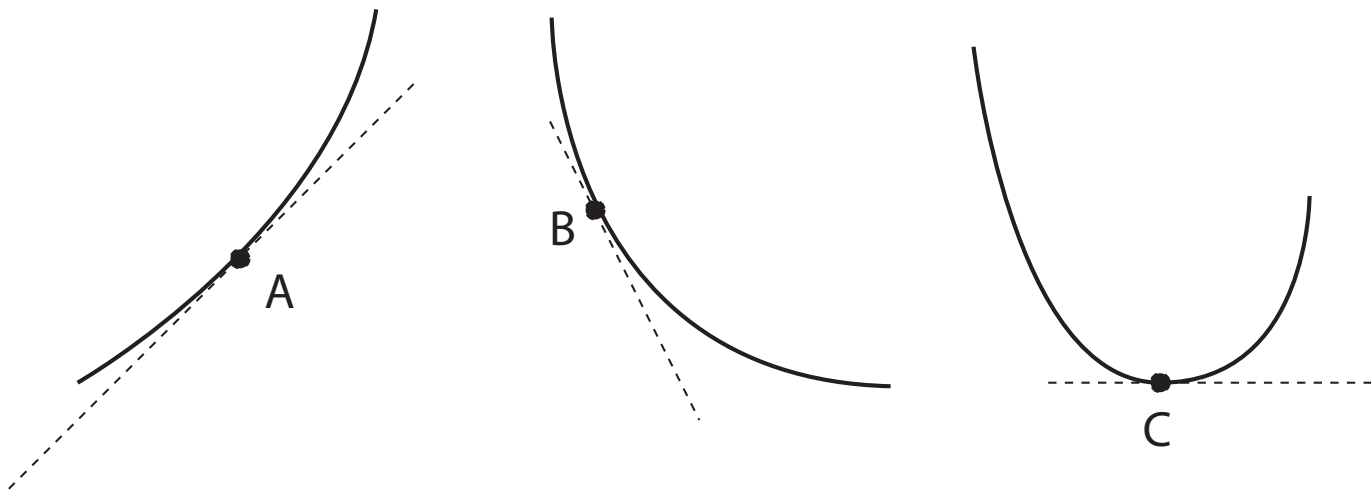
1. Find $f'(x)$
2. Find the set of x such that $f'(x) = 0$
3. Intuitively, if $f'(x)$ changes from negative to positive around x , then x is a minimum; if it changes from positive to negative then x is a maximum; if it does neither then it's a point of inflection
4. Always remember that without more knowledge of f , all we know is that x is a *local* maximum or minimum (or an inflection point); we can't assume that it maximizes or minimizes our function globally

The second derivative

If f' is itself differentiable, then f is *twice differentiable*, and $f'' = \frac{d^2y}{dx^2}$ is the *second derivative*

- Suppose that $f''(x)$ is positive at some point x
- This means that the *slope of the slope* of f is increasing at x
- This means that either:
 - If f is increasing then it's increasing at an increasingly faster rate at x
 - If f is decreasing then it's decreasing at an increasingly slower rate at x
 - If f is flat then it's increasing as x increases
- In all of these cases, f lies above its tangent

- A If f is increasing then it's increasing at an increasingly faster rate at x
- B If f is decreasing then it's decreasing at an increasingly slower rate at x
- C If f is flat then it's increasing as x increases



The second derivative test

- For any critical point, x^* , calculate $f''(x^*)$
 - If $f''(x^*) > 0$ then x^* is a local minimum
 - If $f''(x^*) < 0$ then x^* is a local maximum
 - If $f''(x^*) = 0$ then we don't know what x^* is; we have to examine our function around x^*
- $f'(x) = 0$ is our first-order condition; conditions on f'' are our second-order conditions

Find and classify the critical points of $x^3 - 3x$, and then use this information to plot out the function

Global maxima (minima work the same way)

If a local maximum x^* has the additional property that $f(x^*) \geq f(x)$ for all $x \in \mathbb{R}$ then x^* is a global maximum

Important points about global maxima that we will use shortly

- Suppose that $g(x) = H(f(x))$ where H is a *strictly increasing function*. Then if f has a global maximum at x^* , $g(x)$ also has a global maximum at x^*
 - If $f(x^*) \geq f(x)$ for all x , then $H(f(x^*)) \geq H(f(x))$ for all x , since H is a strictly increasing function
 - It can be useful to take a log; if x^* globally maximizes $\log(f(x))$ then x^* also maximizes $f(x)$
- If x^* is the global max of $f(x)$ then x^* is the global minimum of $-f(x)$

Boundary conditions & asymptotes

If the domain of a function is bounded, it may be the case that a global maximum is attained at one of the boundaries, even when the first-order condition at that point is not satisfied; use common sense and try to have a sense of what your function looks like – some problems are best solved without calculus.

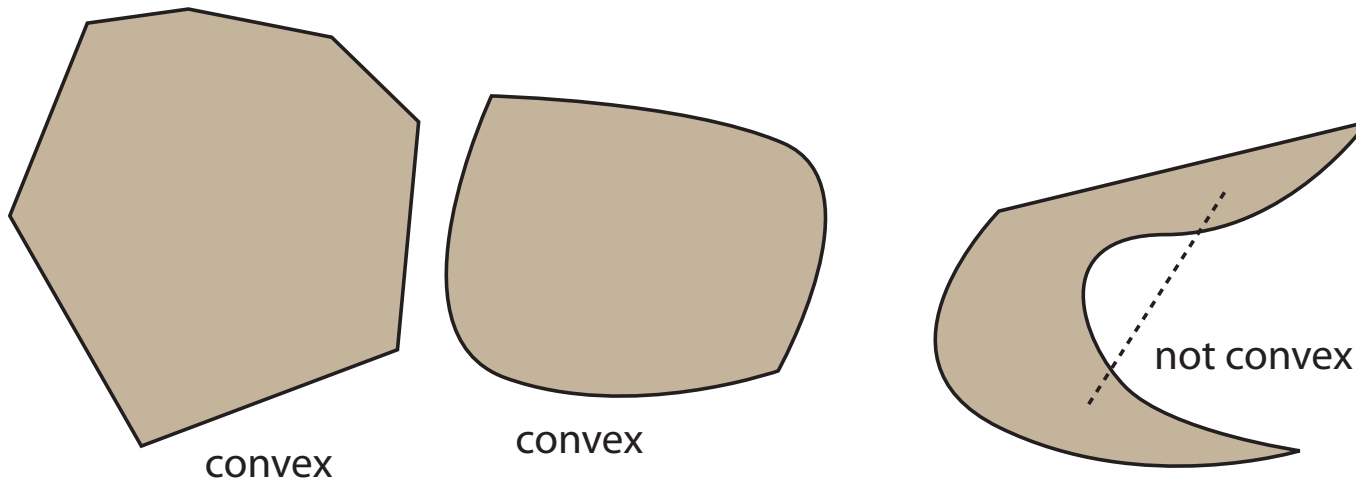
The extreme value theorem: any continuous function on a closed and bounded interval attains a global maximum & minimum

Some functions tend to a straight line for all very large values of $|x|$ and / or $|y|$; it can be useful to know this

- $y = \frac{1}{x}$ ($x > 0$) has a x - and y - asymptotes equal to the axes
- What are the asymptotes for $y = x - \frac{1}{x}$?

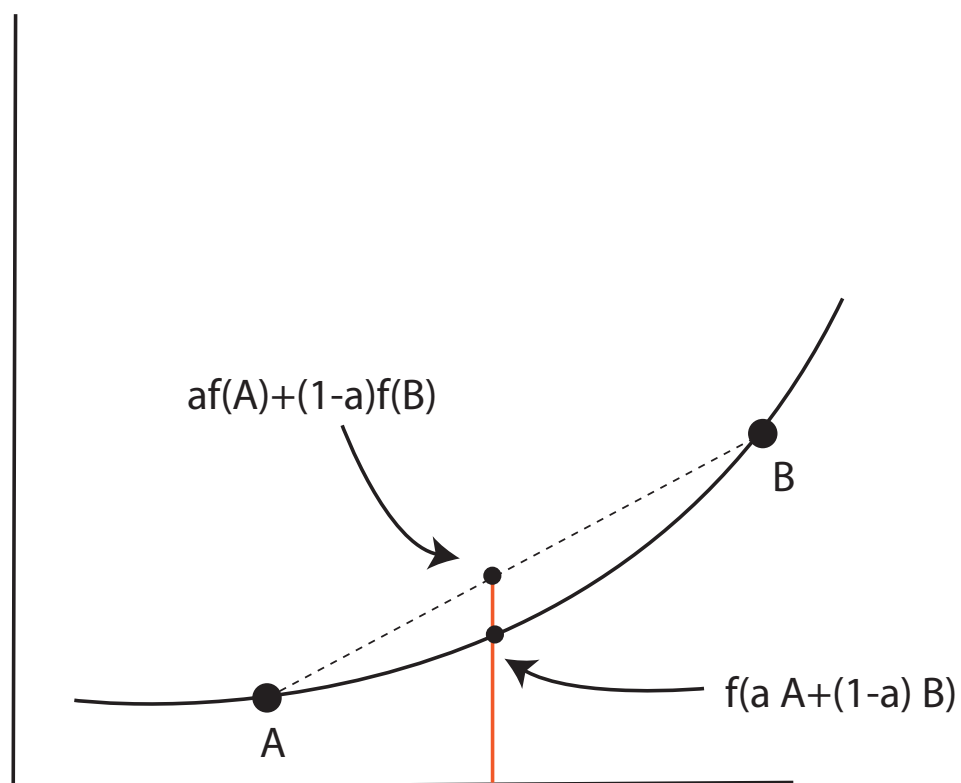
Convexity and concavity

- Convex and concave functions guarantee that local optima are global optima, and are commonly encountered
- A set is *convex* if the line segment connecting any two points in the set is entirely contained within the set
- A function is convex if the set of all points on or above the graph is a convex set



f is a convex function if and only if for all real numbers A, B and all $0 \leq \alpha \leq 1$,

$$\alpha f(A) + (1 - \alpha)f(B) \geq f(\alpha A + (1 - \alpha)B)$$



Convex functions need not be differentiable (e.g. $y = |x|$); the ones we generally encounter are

- If f is twice differentiable, then f is convex if and only if $f'' \geq 0$ for all x
- If f is differentiable then
 - There is either no global minimum, one global minimum, or an interval of global minima
 - There are no local minima that are not global minima
 - Important: if $f'(x^*) = 0$ then x^* is a global minimum
- f is *strictly convex* if $\alpha f(A) + (1 - \alpha)f(B) > f(\alpha A + (1 - \alpha)B)$ when $\alpha \in (0, 1)$

f is *concave* if and only if $-f$ is convex: the set of all points on or below f is convex

- If f is twice differentiable then f is concave if and only if $f'' \leq 0$ for all x
- Important: if f is concave and $f'(x^*) = 0$ then x^* is a global maximum

A simple model of effort; I can exert effort e and receive some benefit $B * e$, but I must pay cost $c * e^2$, with $B, c > 0$

$$U(e) = Be - ce^2$$

- This function is concave: $U' = B - 2ce$ and $U'' = -2c < 0$
- It has a global maximum at $e^* = \frac{B}{2c}$
- *Comparative statics*: given an optimizing point such as this, how does this optimum vary as other parameters of the problem vary?

$$\frac{\partial e^*}{\partial B} = \frac{1}{2c} > 0$$

$$\frac{\partial e^*}{\partial C} = -\frac{B}{2c^2} < 0$$

The exponential function

- The number e is defined by

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.7183$$

- $y = e^x$ is the *exponential function*
- $\frac{d}{dx} e^x = e^x$

Natural logarithms

- The natural logarithm of x is the log to base e of x
- (If $\ln(x) = y$ then $e^y = x$)
- $\frac{d}{dx} \ln(x) = \frac{1}{x}$

Recall your log facts:

- $\ln(x^c) = c \ln x$
- $\ln(xw) = \ln x + \ln w$
- $\ln\left(\frac{x}{w}\right) = \ln x - \ln w$

Application: Optimizing work versus play

Suppose that I have 10 hours in which to either work or play

$$w + p = 10$$

I receive a payoff from any given mixture of these activities that looks like the following:

$$U(w, p) = w^\alpha p^{1-\alpha},$$

for some $\alpha \in (0, 1)$. Notice that if either $w = 0$ or $p = 0$ my utility is zero; some work is required to survive, but without some play *what is the point of living?*

α captures the responsiveness of my well-being to a change in work (holding play constant)

While this looks like a choice over 2 things, it's really a choice over 1:
Letting $p = 10 - w$ we get

$$U(w) = w^\alpha (10 - w)^{1-\alpha},$$

for some $\alpha \in (0, 1)$. What is the w (and consequently, the p that maximizes this?

This is hard to solve explicitly, but we can take a log of this function:

$$U(w) = \alpha \ln(w) + (1 - \alpha) \ln(10 - w)$$

What's the solution? How does it vary as a function of α ? Is utility concave?